

# Capstone Train, Analyze, Improve

Project: Responsible Plant Classifier

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Capstone Dataset and Baseline

Without opening the old lesson, explain one key idea from Capstone Dataset and Baseline. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Audit a difference for Capstone evaluation and revision. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply independently

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

## 6. Solve the given case

Baseline validation accuracy70%; model80%; most remaining errors are dim images.

Plan improvement evidence.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A revision raises overall validation accuracy from 80% to 84% but lowers dim-light accuracy from 60% to 40%. How should the improvement claim change?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 35 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Score the frozen test

Compute the worked candidate metrics and baseline accuracy.

### My reasoning and evidence

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## 13. W2 Audit a difference

What does the one-correct-case improvement establish?

### My reasoning and evidence

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## 14. W3 Label a revision

A threshold changes after test errors are read. Label its score on the same rows.

### My reasoning and evidence

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## 15. W4 Apply independently

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

### My reasoning and evidence

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## 16. W5 Update the card

What must remain visible when version 2 is proposed without a fresh test?

### My reasoning and evidence

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## 17. Feedback, return check and learning record

Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Capstone Train, Analyze, Improve

Choose the next capstone revision from explicit false alarms and misses.

Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.



## Candidate, before revision

|                | predicted spotted | predicted healthy |
|----------------|-------------------|-------------------|
| actual spotted | 4                 | 2                 |
| actual healthy | 1                 | 9                 |

Read a cell using both its row and column labels.

## Capstone Train, Analyze, Improve

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Compare accuracy

Candidate  $13/16=81.25\%$ ; baseline  $10/16=62.5\%$ .



### 2. Compare detection

Candidate catches  $4/6$  spotted leaves; baseline catches  $0/6$ .



### 3. Inspect errors

Review candidate's two misses and one false alarm with conditions and label evidence before revising.

### CHANGE ONE THING

A revision fixes one miss but creates two extra false alarms.

Explain the changed result using the matching lesson.

Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## 20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose the next capstone revision from explicit false alarms and misses.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Review candidate's two misses and one false alarm with conditions and label evidence before revising.

1. Inspect errors
2. Compare accuracy
3. Compare detection

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect errors Review candidate's two misses and one false alarm with conditions and label evidence before revising. Read the picture in order. Review candidate's two misses and one false alarm with conditions and label evidence before revising. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### Practice 2: explain the deciding evidence

Changed-case practice: A revision fixes one miss but creates two extra false alarms. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. TP 5, FN 1, FP 3, TN 7 gives  $12/16=75\%$  accuracy and  $5/6$  recall.
3. This one example guarantees correct results for every future input.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: TP 5, FN 1, FP 3, TN 7 gives  $12/16=75\%$  accuracy and  $5/6$  recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

### Repair a missing building block

Review Capstone Dataset and Baseline (ai-4/34). Goal: Define a feasible capstone question and acceptance criteria. Short practice: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What evidence contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at prediction time. Short practice: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### What I repaired, the evidence I kept and the next test

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